Telecom Reliability, Availability, and Maintenance and the Effects on Churn and Loyalty

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Introduction

The telecommunications industry in the United States straddles the boundary between public utility and privatized enterprise, raising important questions about service expectations and market behavior (Federal Communications Commission [FCC] v. Consumers' Research, 2022). Nearly every individual in the U.S. interacts with telecom services daily—at work, at school, at home, or on mobile devices—making performance a vital component of customer experience (CX) (Gelles-Watnick, 2024). Central to any discussion on Telecommunication CX is Reliability, Availability, and Maintainability. Reliability, Availability, and Maintainability (RAM) is a framework, originally developed by NASA for mission-critical systems (NASA, 1998; Ebeling, 1996). While RAM is mandated in public utilities, its role in customer satisfaction and competitive differentiation in privatized telecom remains unclear (FERC, n.d; Wetzel et al., 2024).

The problem this study addresses is that, despite the competitive nature of telecom services, it is unknown whether RAM is the most influential factor in reducing customer churn and building loyalty, or whether other lifecycle drivers (such as pricing, customer service, or bundling) have a greater impact. The purpose of this research is to explore the relative importance of RAM in the customer lifecycle of privatized telecommunications providers and to identify whether RAM can be used as a key strategic differentiator in reducing churn and enhancing customer loyalty.

Research questions include:

RQ1: What is the typical customer lifecycle in the U.S. telecommunications industry?

RQ2: What factors are most predictive of customer churn or loyalty?

RQ3: What is RAM and how is it applied in telecommunications?

RQ4: To what extent does RAM affect customer loyalty and reduce churn?

RQ5: Is RAM the most critical factor, or are other variables more significant?

This study contributes to literature by extending the application of the RAM framework into the domain of competitive telecommunications markets. By linking engineering reliability concepts to business outcomes like churn, this research helps fill a gap at the intersection of service engineering and customer experience strategy.

Literature Review

Customer Lifecycle Management

To address Research Question 1— *"What is the typical customer lifecycle in the U.S. telecommunications industry?"*—we must first define the stages of a customer's journey in this sector. Flu (2014) proposed a ten-stage lifecycle model specific to telecom: Aware, Interest, Decide, Buy, Early Use, Regular Use, Change of Use, Incident, Reconsider, and Leave. Alternatively, Zuhyle and Gunaratne (n.d.) outlined a customer lifecycle used in South Asian markets such as Sri Lanka and Bangladesh, including Targeting, Enquiry Management, Welcoming, Getting to Know the Customer, Customer Development, Managing Problems, and Winback. Their model avoids assumptions commonly found in Western, postpaid-dominant environments—such as contract-driven loyalty—and instead emphasizes operational integration with key business metrics like ARPU (Average Revenue Per User), value-added services, and knowledge sharing. ServiceNow (2025) offers a more marketing-centric approach, identifying five streamlined stages: Awareness, Acquisition, Conversion, Retention, and Advocacy.

Building on the work of Flu (2014), Zuhyle & Gunaratne (n.d.), and ServiceNow (2025), this unified model consolidates overlapping stages. A unified nine-stage telecom customer lifecycle can be proposed. Notably, customer usage is treated as a distinct phase with critical sub-stages—initial use, early use, regular use, and long-term use—during which behavioral insights and RAM (Reliability, Availability, Maintainability) metrics are most observable:

1. Targeting and Awareness.
2. Interest, Consideration, and Enquiry Management.
3. Decision.
4. Acquisition, Welcoming, and Onboarding.
5. Usage. (Get to know the customer).
    - *Initial Use. 0-30 days.*
    - *Early Use. 30-90 days.*
    - *Regular Use. 90 days until contract completion.*
    - *Long term Use. Beyond the contract.*
6. Customer Development, Change of Use, and Conversion.
7. Incident and Problem Management.
8. Reconsideration and Retention.
9. Advocacy and Winback.

This nine-phase structure supports lifecycle-driven performance monitoring, particularly during early engagement and incident resolution stages. For example, early and regular use phases align with RAM-informed quality expectations, including First Call Resolution (FCR), Mean Time to Resolve (MTTR), and Service Level Agreement (SLA) compliance.

Causes and Predictors of Customer Churn in Telecommunications

Customer churn in telecommunications is a critical concern, often driven by both technical performance and behavioral perceptions. Ghotbi, Dhar, and Weinberg (2021) demonstrated how consumer behavior can be modeled using econometric techniques, offering a rigorous analytical approach to segmenting customers and identifying churn risk based on behavioral patterns and demographic variables. These methods are particularly applicable to telecommunications, where customer decisions often follow observable usage and experience-based trends.

Ookla (2021) identified several leading indicators of Quality of Experience (QoE) that predict churn, including web page load times, download and upload speeds, video start times, latency, jitter, and NetFlow anomalies. Their research also emphasizes the role of early infrastructure investment (EII) in sustaining performance expectations and reducing churn.

Wagh, Andhale, Pansare, Ambadeka, and Gawande (n.d.) identified network-related issues—such as poor coverage, frequent call drops, and signal inconsistencies—as primary contributors to customer dissatisfaction and subsequent churn. They further highlighted pricing structures, customer support quality, switching costs, and customer demographics as influential churn factors. Notably, their research points to the predictive value of analyzing customer complaints and signal errors through social media and network topology data, underscoring the importance of peer influence and public sentiment.

These findings demonstrate that churn is not only a product of direct service failure but also of perceived value, ease of switching, and social signaling. Thus, churn modeling in telecom

must integrate technical, behavioral, and social indicators to accurately predict and mitigate customer loss.

Technical background on RAM

The concepts of reliability, availability, and maintainability (RAM) originated in the defense and aerospace sectors, where system dependability was paramount. The U.S. Department of Defense first formalized RAM principles in MIL-HDBK-217, published in 1965. This handbook introduced quantitative models for assessing system reliability through metrics such as Mean Time Between Failure (MTBF), Mean Time to Repair (MTTR), and Availability (A), which can be calculated as per Appendix 2: Reliability Calculations. These metrics form the basis for evaluating electronic and communications equipment used in mission-critical military systems.

Building upon these foundations, the Bell System Technical Journal (BSTJ), originally launched by AT&T in 1922, contributed significant research on switching system reliability, network continuity, and service-level performance in public telecommunications networks. This work laid the groundwork for industry-level understanding of availability engineering across distributed systems.

In the 1990s, RAM concepts expanded into systems engineering and commercial logistics domains. Blanchard and Fabrycky (1990) emphasized the tradeoffs between MTBF, MTTR, and system lifecycle cost in Systems Engineering and Analysis**,** introducing RAM principles into the broader context of service delivery and resource optimization. Ebeling (1996), in *An Introduction to Reliability and Maintainability Engineering*, made these military and industrial models accessible to academic and civilian engineers by presenting exponential failure distributions,

component reliability, and system-level modeling using reliability block diagrams. This work also helped formalize the acronym “RAM” in both scholarly and professional practice.

In 1998, NASA published its Reliability-Centered Maintenance Guide for Facilities and Collateral Equipment**,** applying RAM methodologies to ensure uptime in infrastructure-critical systems. NASA utilized concepts of Fault Tree Analysis (FTA), Maintenance Task Analysis (MTA), and Problems/Faults/Anomalies (P/F/A) Management to minimize troubleshooting. This document further extended RAM's applicability from defense to civilian and enterprise environments.

The International Telecommunication Union (ITU), a key global standards body, institutionalized RAM-related metrics through its *E.800 series* of Quality of Service (QoS) recommendations (ITU, 2008). These standards define reliability in terms of availability, mean downtime, and service restoration expectations, establishing baseline expectations for international telecom operators. According to the ITU, customer satisfaction is divided into Technical QoS and Non-Technical QoS. QoS Technical is divided into network performance and terminal (Personal Computer or Server) performance. Non-technical QoS is made of the Point of Sale and Customer Care.

Together, these works provide a robust technical foundation for applying RAM in the telecommunications industry, where uptime, failure frequency, and repair responsiveness directly impact service continuity, customer satisfaction, and regulatory compliance.

RAM Metrics Across the Customer Lifecycle

The "Incident and Problem Management" stage is strongly influenced by measurable RAM indicators that directly affect customer satisfaction, perceived service quality, and churn risk. These indicators can be found in Appendix 1: Key Performance Indicators. These metrics align with expectations set during onboarding and usage phases and form the operational core of churn prediction systems and SLA compliance in the telecommunications sector.

The Role of RAM in Churn and Customer Loyalty

Reliability, availability, and maintainability (RAM) metrics have a direct and measurable impact on customer satisfaction, retention, and loyalty within the telecommunications industry. These indicators influence key phases of the customer lifecycle—particularly usage, incident management, and reconsideration—by shaping customer perceptions of service quality and dependability.

According to J.D. Power's (2023) Telecommunications Intelligence Report, consistent problem resolution and high-quality customer care are primary drivers of loyalty in broadband and streaming services. The report found that 88% of TV streamers were unlikely to switch providers when technical issues were handled effectively. For internet service, customers expressed strong preferences for seamless, uninterrupted connections with fast download speeds and stable signals. Increasing reliance on mobile and streaming data, particularly among younger consumers, has decreased customer tolerance for buffering, slow page loads, or network interruptions.

RAM-related performance indicators, including First Call Resolution (FCR), Mean Time to Resolve (MTTR), MTBF, and service restoration speed—are increasingly used as proxies for perceived quality. These indicators are also reflected in Net Promoter Scores (NPS), which J.D. Power indices correlate with satisfaction and brand loyalty. As such, RAM performance serves not only operational goals but also as a predictive input into marketing and retention models.

Generational differences also affect the perception of RAM. J.D. Power (2023) observed that younger users were more likely to churn in response to performance lags or slow speeds, while older users prioritized price-to-value ratios. Across all demographics, the perceived value of the service was strongly linked to network reliability and responsiveness—two attributes driven directly by RAM metrics.

Internationally, similar findings emerge. OFCOM (2022), the United Kingdom's telecommunications regulator, reported that customer satisfaction levels were highly correlated with service reliability. Lower satisfaction was linked to perceived reductions in signal quality, speeds, and complaint resolution. Social influence also played a role—telecom providers with poor RAM outcomes saw fewer peer recommendations and more complaints on social media, resulting in reputational damage and higher churn rates.

Behavioral and contractual factors also play a significant role in churn. Ahn, Han, and Lee (2006) investigated determinants of customer churn in the Korean mobile telecommunications market and identified several non-technical factors that contribute to switching behavior:

1. Price-Based Switching: Customers often migrate to competitors offering lower prices or promotional deals, although retention hinges on value beyond pricing alone.

2. End-of-Contract Churn**:** Switching frequently occurs at the conclusion of service agreements. High switching costs (e.g., early termination fees or loss of bundled services) mitigate churn, while minimal switching barriers accelerate it.
3. Poor Customer Experience: Dissatisfaction resulting from product performance issues, unaddressed complaints, and technical failures is a strong predictor of churn.
4. Low Brand Loyalty and Customer Inertia: In low-differentiation markets, customers may churn due to apathy or lack of emotional attachment to the brand. Conversely, loyalty programs and complex switching procedures can inhibit churn.
5. Demographic Effects: Younger and more mobile users exhibit higher churn rates due to their lower switching resistance and demand for performance.
6. Competitive Market Activity: Aggressive marketing, bundling strategies, and new features by rival firms often provoke churn.
7. Technological Advancement: Customers may switch service providers in pursuit of superior technologies.
8. Declining Perceived Value: Even in the absence of price increases, perceived reductions in service quality or features can raise churn risk.

In a related study Kim and Jeong (2004) found that satisfaction—driven by service quality, price fairness, network reliability, and customer support—significantly predicted customer retention. Importantly, low switching barriers like simple porting procedures and no financial penalties were linked to higher churn rates when satisfaction was mediocre.

At the regulatory level, the significance of RAM is reinforced by the Federal Communications Commission (FCC), which mandates reporting outages lasting more than 30

minutes, or any 911 outage, to the Cybersecurity and Communications Reliability Division. These reports inform the industry of the best practices and reliability standards (FCC, 2023). Similarly, the United Kingdom's OFCOM (2022) found that satisfaction across mobile, broadband, and landline services was strongly tied to performance consistency and complaint resolution. Poor RAM performance corresponded with fewer peer recommendations, higher social media complaints, and ultimately, increased churn.

Together, these findings illustrate that managing churn and loyalty requires a dual focus: on RAM performance indicators that influence perceived service quality, and on behavioral and contextual factors that influence switching behavior. Integrating both domains into customer lifecycle management strategies provides a more complete model for anticipating and reducing customer loss in the telecommunications sector.

Research Methodology

Research questions 1, 2, and 3 can be thoroughly answered through literature review. Research question 4 *To what extent does RAM affect customer loyalty and reduce churn?* and question 5 *Is RAM the most critical factor or are other variables more significant?* begins understanding through literature review, but only to define what can affect loyalty and reduce churn. To understand how RAM affects customer loyalty we must use sequential explanatory design. This approach starts with quantitative data collection and analysis to identify patterns, followed by qualitative inquiry to contextualize and explain those findings. This method is appropriate because RAM metrics are inherently quantitative. Customer loyalty and churn involve human perceptions, best explored qualitatively. Prior studies by Ghotbi, Dahr, & Winberg (2021) have shown the value of econometric modeling followed by interpretive policy

implications. This is also in line with experimental advertising studies such as Steenis et al. (2023), which used surveys and experimental design to gauge perception-based constructs.

Step-by-Step Execution

Phase 1: Quantitative Analysis

- Objective: Determine statistical relationships between RAM metrics and customer churn rates.
- Method: Prepare a structured survey sent to telecom customers (or internal network reliability logs cross-referenced with customer behavior). Compare surveys to RAM metrics.
- Variables:
    - Independent: Refer to RAM Metrics Across the Customer Lifecycle. Examples are MTBF, MTTR, network uptime, service restoration time.
    - Dependent: Customer churn rate, loyalty indices (e.g., MOS, NPS).
- Sample: Minimum of 385 responses (for 95% confidence, ±5% margin of error, population >100,000).
- Analysis: Logistic regression or Structural Equation Modeling (SEM) (Homer et al., 2013; Kline, 2015).
    - Example from Steenis et al.: ANOVA and mediation modeling via SPSS Process Macro (Steenis et al, 2023).
- Tools: Qualtrics/SurveyMonkey for data collection, SPSS or R for analysis.

Phase 2: Qualitative Analysis

- Objective: Explore customer perceptions of RAM impact on service satisfaction and switching behavior.
- Method: Semi-structured interviews or focus groups with a stratified sample.
- Questions: Explore themes such as:
    - "How do you define a reliable telecom service?"
    - "Have you ever switched providers due to network issues? What triggered that?"
- Analysis: Thematic analysis using NVivo or manual coding (Braun & Clarke, 2006; Blakely & Jackson, 2013).
- Support: Literature from telecom customer behavior studies and decision psychology.

Data Types to Collect

- Quantitative:
    - RAM performance data.
    - Survey responses on service satisfaction.
    - Switching history.
- Qualitative:
    - Interview transcripts.
    - Customer support feedback logs (if available).

Significance of this Research

The primary objective of this research is to evaluate the extent to which Reliability, Availability, and Maintainability (RAM) metrics influence customer churn and foster customer loyalty in the telecommunications sector. By identifying which specific RAM dimensions have the strongest correlations with customer experience indicators such as Mean Opinion Score (MOS) and Net Promoter Score (NPS), this study aims to inform data-driven strategies for service improvement and retention. A secondary objective is to assess whether RAM metrics are the most significant predictors of MOS and NPS, or if other performance or experiential metrics exert greater influence on customer perceptions and loyalty behavior.

Implications

This research contributes to telecom churn expanding on current literature by introducing performance metrics (RAM) as potential predictors of customer loyalty or churn. This is relatively under-explored domain in marketing and service management theory. If RAM metrics significantly impact Mean Opinion Score (MOS) or Net Promoter Score (NPS), which in turn appear to correlate with churn and loyalty, then findings would support a mediation model that links service quality to customer lifecycle.

This work is cross-disciplinary which ties engineering metrics from RAM with marketing constructs of churn, loyalty, and satisfaction. Empirically, this study seeks to validate that RAM metrics are statistically significant indicators of churn, loyalty, and satisfaction. It will also determine if RAM proves to be less influential than other factors such as price and customer service. This will guide future measurement priorities. Management priorities including network

investment prioritization or churn reduction strategies. NPS and MOS can be used as benchmarks to act as an early warning system for network-induced churn. If RAM is confirmed to significantly affect churn this may lead to tighter regulatory insight of SLA enforcement, particularly in underserved and rural areas.

This study may also lead to expanded reporting of certain RAM statistics like how utilities report outages. Currently the FCC is requiring telecoms to provide an FCC consumer label on services which look like nutrition labels on food consumables. Using longitudinal data or time-series forecasting, it also informs how technical reliability evolves over in time in relation to customer behavior. The results can serve as benchmarks for comparing performance across providers or as inputs for Strength, Weakness, Opportunity, Threats (SWOT) or balanced scorecard assessments. The results may show the return on investment (ROI) in predictive maintenance, AI-driven fault detection, or customer communication systems.

Appendix 1: Key Performance Metrics.

Key performance metrics include:

- Change Management (CM) and Planned Notification Handling.
- Customer Support/Care Metrics:
    - Average Wait Time (AWT).
    - Repeat Calls.
    - First Call Resolution (FCR).
    - Average Call Duration (ACD).
    - Escalation Response Time (ERT).
    - Percent Abandoned Calls.
- Resolution and Recovery Metrics:
    - Mean Down Time (MDT).
    - Average Time to Resolve (ATTR).
    - Mean Time to First Failure (MTTFF).
    - Mean Time to Failure (MTTF).
    - Mean Time to Resolve (MTTR).
    - Mean Time Between Replacements.
    - Mean Time Between Interruptions (MTBI).
    - Mean Time Between Failures (MTBF).
- Service Delivery Metrics:
    - Bit Error Ratio (BER).
    - Frame Loss Variation (FLV).

    - Frame Delay Variation (FDV).
    - Packet Loss.
    - IP Packet Error Ratio (IPER).
    - IP Packet Loss Ratio (IPLR).
    - Percent IP service availability (PIA).
    - Packet Transfer Delay (PTD or Jitter).
    - End-to-end delay (latency).
    - Uptime-to-Downtime Ratio.
    - Speed Attained vs. Advertised.
    - Service Level Agreement (SLA) Fulfillment.
    - Service Quality Agreement (SQA) Fulfillment.
- Quality Assessments:
    - Quality of Service (QoS).
        - Quality of Service Required (QOSR).
        - Quality of Service Offered (QOSO).
        - Quality of Service Delivered (QOSD).
        - Quality of Service Experienced (QOSE).
    - Quality of Experience (QoE).
    - Mean Opinion Score (MOS).
    - Net Promoter Score (NPS).

Appendix 2: Reliability Calculations

Availability is calculated as:

$$\boldsymbol{A = \frac{MTBF}{MTBF + MTTR}}$$

MTBF = Uptime / Number of System Failures.

MTTR = CM Downtime / Number of System Failures.

This is also known as inherent availability or A_i. For a single system this is defined using Mean Time to Failure (MTTF):

$$\boldsymbol{A_i = \frac{MTTF}{MTTF + MTTR}}$$

Below is operational availability, $\boldsymbol{A_o}$.

$$\boldsymbol{A_o = \frac{Uptime}{Uptime + Downtime}}$$

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